Parul® University
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NAAC GRADE A++

Information and Communication Technology

Introduction to Microprocessors

Study Guide

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1.1 Overview of Microprocessors

A microprocessor is an integrated circuit (IC) that contains the entire central processing unit (CPU) of a computer on a single chip. It carries out instructions from programs by performing basic arithmetic, logic, control, and input/output operations.

Key Features:

- Small and compact: It fits the processing power of an entire CPU onto a tiny chip.
- Programmable: It can run different software programs.
- Fast: Executes millions (or billions) of instructions per second.
- Made of millions (or billions) of transistors: Transistors are tiny electronic switches that control the flow of electricity.

How it works:

- Fetch: The microprocessor retrieves an instruction from memory.
- Decode: It interprets the instruction.
- Execute: It performs the instruction (like adding numbers, moving data, etc.).
- Store: It writes back the result to memory if needed.

Architecture

A microprocessor's architecture defines how it is designed internally. The two main parts are:

- Control Unit (CU): Directs the operation of the processor. It tells the memory, ALU, and I/O devices how to respond to program instructions.
- Arithmetic Logic Unit (ALU): Performs arithmetic operations (like addition, subtraction) and logic operations (like AND, OR, NOT).
- Registers: Small, fast storage locations inside the CPU used to hold temporary data and instructions. Examples: Accumulator, Instruction Register, Stack Pointer.
- Bus System: Pathways for transferring data:
    - Data bus (carries data)
    - Address bus (carries memory addresses)
    - Control bus (carries control signals)
- Clock Speed
    - Microprocessors are synchronized by a clock signal.
    - Clock speed is measured in Hertz (Hz) — typically GHz (billions of cycles per second).
    - One clock cycle = fetch + decode + execute one simple instruction (depends on design, some instructions may need multiple cycles).
- Instruction Set Architecture (ISA)
    - ISA defines the set of instructions a microprocessor understands (machine language).

  - Examples:
    - CISC (Complex Instruction Set Computing): (e.g., Intel x86) — many complex instructions.
    - RISC (Reduced Instruction Set Computing): (e.g., ARM) — fewer, simpler instructions for faster execution.
- Caches:
  - Small, very fast memory inside or close to the CPU (L1, L2, L3 caches).
  - Stores frequently used data and instructions to reduce access time from slower main memory (RAM).
- Interrupt System
  - Microprocessors can respond to asynchronous events using interrupts.
  - When an interrupt occurs (like input from a keyboard), the CPU pauses its current task, handles the event, and then resumes.

Component	**Role**
ALU	Performs calculations and logical operations
CU	Controls instruction execution
Registers	Temporary data storage
Bus System	Transfers data and control signals
Cache	Fast memory to speed up processing
Clock	Synchronizes operations
Cores	Multiple processing units for parallel tasks

1.2 History and Evolution

1st Generation (1971–1973): 4-bit Microprocessors

- Notable Chip: Intel 4004
    - Year: 1971
    - Bit Size: 4-bitTransistors: ~2,300
    - Clock Speed: 740 KHz
    - Technology: PMOS
    - Use: Basic calculators and simple embedded systems
- Significance:
    - First commercially available microprocessor
    - Designed for a Japanese calculator company (Busicom)
    - Could perform simple arithmetic and logic operations

2nd Generation (1974–1978): 8-bit Microprocessors

- Notable Chips:
    - Intel 8080 (1974)
    - Intel 8085 (1976)
    - Motorola 6800, Zilog Z80
- Improvements:
    - Higher clock speed (~1 MHz)
    - More instructions and addressing modes
    - Used NMOS technology
    - Supported interrupts and I/O
- Applications:
    - Early computers (like the Altair 8800)

3rd Generation (1978–1982): 16-bit Microprocessors

- Notable Chips:
    - Intel 8086 (1978) – the basis of x86 architecture
    - Intel 80186, 80286
    - Motorola 68000
- Key Features:
    - Addressing of more memory (up to 1 MB and beyond)
    - Segment-based memory management (Intel)
    - Faster execution with pipeliningUsed in IBM PCs (8088 variant)

4th Generation (1982–1990): 32-bit Microprocessors

- Notable Chips:
    - Intel 80386 (1985)
    - Intel 80486 (1989)
    - Motorola 68020/68030

- Advancements:
  - 32-bit internal and external data bus
  - On-chip cache memory
  - Pipelining and integrated FPU (80486)
  - Virtual memory and multitasking support

5th Generation (1990s):

- Superscalar & Early 64-bit
- Notable Chips:
  - Intel Pentium (1993)
  - AMD K5, K6
  - PowerPC, MIPS, SPARC
- Key Innovations:
  - Superscalar architecture: multiple instructions per clock
  - MMX technology (for multimedia)
  - Integrated L2 cache in some models
  - Introduction of parallel instruction pipelines

6th Generation (2000s): 64-bit, Multi-Core

- Notable Chips:
  - Intel Core, Pentium 4, Athlon 64, PowerPC G5
- Milestones:
  - 64-bit processing becomes standard
  - Multicore processors emerge (dual-core, quad-core, etc.)
  - Enhanced power management
  - Introduction of virtualization support

Modern Era (2010–Present): Multicore, AI, Mobile, ARM

- Key Players:
  - Intel Core i3/i5/i7/i9, AMD Ryzen, Apple M1/M2/M3
  - ARM Cortex-A, Qualcomm Snapdragon, Apple Silicon
- Technologies:
  - 7nm and 5nm fabrication nodes
  - 10+ core processors (desktop/server)
  - Integrated GPUs
  - AI accelerators and neural engines
  - Ultra-low-power mobile processors
  - System-on-Chip (SoC) integration
- Trends:
- Mobile & Edge computing (phones, tablets, IoT)
- Parallel computing & GPUs for AI
- Custom microarchitectures (e.g., Apple's ARM-based M-series)

- Green computing and thermal efficiency

1.3 Architecture of a Typical Microprocessor

Arithmetic and Logic Unit (ALU)

- Function: Performs all arithmetic operations (add, subtract, multiply, divide) and logical operations (AND, OR, NOT, XOR).
- Importance: It's the heart of the processor for computation.
- Subcomponents: Accumulators (temporary registers for operations)Flags (carry, zero, sign, overflow indicators)

Control Unit (CU)

- Function: Directs the operation of the processor.
- Tasks:
    - Decodes instructions fetched from memory.
    - Coordinates between ALU, memory, and I/O devices.
    - Generates control signals to manage data flow.
- Instruction Decoder: Part of CU that translates binary instructions into control signals.

Registers: Registers are small, fast storage units within the CPU used to hold data and instructions temporarily.

- General Purpose Registers (GPRs): Hold intermediate data (e.g., AX, BX in x86).
- Special Purpose Registers:
    - Program Counter (PC): Holds the address of the next instruction.
    - Instruction Register (IR): Holds the current instruction.
    - Stack Pointer (SP): Points to the top of the stack.
    - Status Register/Flag Register: Stores the result of operations (e.g., Zero flag, Carry flag).

Cache Memory

- Function: High-speed memory located inside or close to the processor to store frequently accessed data.
- Levels:
    - L1 Cache: Fastest, smallest, closest to the core.
    - L2 Cache: Slower than L1 but larger.
    - L3 Cache: Shared among cores in multicore processors.

System Buses: Buses are pathways that transfer data and signals between microprocessor components and other parts of the computer.

Types:

- Data Bus: Transfers actual data.
- Address Bus: Carries memory addresses.
- Control Bus: Carries control signals like read/write.

Clock Generator:

- Provides timing signals to synchronize all operations.
- Measured in MHz or GHz (e.g., 3.2 GHz = 3.2 billion cycles per second).

Input/Output (I/O) Ports:

- Allow communication with external devices (keyboard, display, etc.).Mapped I/O and
- Memory-mapped I/O are two common methods for handling I/O operations.

Component	Function/Tasks	Importance/Notes	Subcomponents/Details
Arithmetic and Logic Unit (ALU)	Performs arithmetic (add, subtract, multiply, divide) and logical operations (AND, OR, NOT, XOR)	Core unit for computation	- Accumulators (temporary storage) - Flags (carry, zero, sign, overflow)
Control Unit (CU)	Directs processor operations; decodes instructions; coordinates ALU, memory, I/O	Manages data flow and instruction execution	- Instruction Decoder (translates binary instructions)
Registers	Temporary, fast storage for data and instructions	Enables quick access to data during operations	- General Purpose Registers (GPRs) - Special Purpose Registers: Program Counter (PC), Instruction Register (IR), Stack Pointer (SP), Status/Flag Register
Cache Memory	Stores frequently accessed data for faster CPU access	Reduces access time to main memory	- L1 Cache (smallest, fastest) - L2 Cache (larger, slower) - L3 Cache (shared among cores)
System Buses	Transfers data, addresses, and control signals within the system	Links CPU to memory and I/O devices	- Data Bus (data transfer) - Address Bus (address transfer) - Control Bus (control signal transfer)
Clock Generator	Provides timing signals to synchronize processor operations	Controls the pace of instruction execution	- Frequency measured in MHz or GHz (e.g., 3.2 GHz)
Input/Output (I/O) Ports	Interface for external devices (e.g., keyboard, display)	Enables communication with external world	- Mapped I/O - Memory-mapped I/O

1.4 Microprocessor operations: Fetch, Decode, and Execute

Microprocessor operations follow a structured cycle known as the Fetch-Decode-Execute cycle, which is the core process by which a microprocessor executes instructions from a program

Fetch: This is the stage where the microprocessor retrieves the next instruction to be executed from memory.

- Program Counter (PC): Holds the address of the next instruction.
- Memory Address Register (MAR): The address from the PC is copied here to access memory.
- Memory Data Register (MDR): The instruction fetched from memory is temporarily held here.
- Instruction Register (IR): The fetched instruction is moved here for decoding and execution.

Decode: At this stage, the Control Unit interprets the instruction stored in the Instruction Register.

- Opcode (operation code) and operands are extracted from the instruction.
- The control logic interprets what action is to be taken.
- If needed, it determines the addressing mode (immediate, direct, indirect, etc.).

Execute: The microprocessor carries out the decoded instruction.

- Arithmetic/logic operations are performed by the Arithmetic Logic Unit (ALU).
- Data transfer operations involve moving data between registers or memory.
- Control operations may involve jumping to a different instruction address.

Aspect	**Fetch**	**Decode**	**Execute**
Purpose	Retrieve instruction from memory	Interpret the fetched instruction	Perform the operation specified
Main Component Involved	Program Counter (PC), Memory, Instruction Register (IR)	Control Unit	ALU (Arithmetic Logic Unit), Registers, Control Unit
Input	Address from Program Counter (PC)	Instruction in Instruction Register (IR)	Decoded signals and operands
Output	Loaded instruction in Instruction Register (IR)	Control signals for execution	Result of the instruction execution
Process	PC sends address → Memory sends instruction → IR stores it	Control unit analyzes opcode and operands	ALU or other parts carry out operation, store result
Time Taken	Typically 1 machine cycle	Typically 1 machine cycle	Varies (may take more cycles if complex)
Memory Interaction	Yes (read from memory)	No	Maybe (for read/write operations)

Aspect	Fetch	Decode	Execute
Typical Operations	Read instruction	Identify operation and operands	Add, subtract, move, jump, compare, etc.
Key Registers Used	PC, MAR, MDR, IR	IR, Control Unit	ALU, Accumulator, General-purpose registers
Result	Instruction ready to be processed	Control signals generated for execution	Desired computation or data manipulation completed

1.5 Microprocessor vs. Microcontroller

Feature / Operation	Microprocessor	Microcontroller
Definition	CPU on a single chip	CPU + memory + peripherals on one chip
Main Focus	Processing power	Task-specific control operations
Memory (RAM, ROM)	External (separate chips required)	Internal (built-in RAM, ROM/Flash)
I/O Ports	External (needs support chips)	Internal (built-in)
Operation Speed	Generally higher (faster CPUs)	Moderate (optimized for control tasks)
Complexity of Design	Complex (requires additional circuits)	Simple (integrated all-in-one)
Applications	Computers, servers, high-end tasks	Embedded systems (e.g., washing machines, IoT devices)
Power Consumption	Higher	Lower (optimized for energy efficiency)
Cost	More expensive overall (including external parts)	Cheaper (self-contained)
Real-Time Operations	Not ideal without RTOS (Real-Time OS)	Good for real-time applications
Examples	Intel Core i7, AMD Ryzen, ARM Cortex-A	Arduino (ATmega328P), STM32, ESP32

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